

Suite2p Cell Classification — Feature Reference Guide

Detailed explanation of all spatial, temporal, and biological features used by the classifiers (Continuous, No-Idx, and 3-Bin-Idx configurations).

Feature Name	Category	Calculation / Formula	Utility & Importance	Meaning & Comments
number_of_bright_pixels	Spatial	<code>sum(lam > 0.1 * max(lam))</code>	Highly Useful	Measures spatial footprint compactness. Real cell somas are highly concentrated and have few pixels with high weights (lam). Blood vessels or out-of-focus background are diffuse and yield much larger bright pixel counts.
solidity	Spatial	<code>s.get('solidity', 1.0)</code>	Useful	Ratio of the ROI area to its convex hull. Neuronal somas are convex (solidity near 1.0). Elongated, fragmented, or hollow footprints represent artifacts and yield low solidity.
mrs	Spatial	<code>s.get('mrs', 0)</code>	Moderate	Mean square distance of pixels to the center of mass. Represents compactness. Compact circular structures have lower Mrs values.
roi_idx_norm	Spatial (Continuous)	<code>idx / n_rois</code>	Overfits	Continuous normalized sorting rank. Represents Suite2p's internal cell probability order. Highly predictive but prone to severe overfitting on noisy runs.
roi_idx_norm_3bin	Spatial (Binned)	<code>digitize(idx / n_rois, [0.1, 0.4])</code>	Highly Useful	Coarse discretized sorting rank (0, 1, or 2). Limits index feature power (plunging importance rank to #20), forcing the model to prioritize biology while retaining a weak prior.
skew_f	Trace Stats	<code>skew(F)</code>	Highly Useful	Skewness of raw trace. Real calcium activity has asymmetric upward transients and slow decays (high positive skew). Background noise is symmetric (skew near 0).
std_f	Trace Stats	<code>std(F)</code>	Highly Useful	Standard deviation of raw trace. Measures absolute variance. Active somas show high variance; inactive ROIs or pure noise show very low variance.
max_to_mean_f	Trace Stats	<code>max(F) / mean(F)</code>	Useful	Maximum to mean ratio of raw trace. Helps identify cells with sparse, high-amplitude calcium transients.
cv_f	Trace Stats	<code>std(F) / mean(F)</code>	Moderate	Coefficient of variation of raw trace. Normalized variance metric.
skew_fneu	Trace Stats	<code>skew(Fneu)</code>	Moderate	Skewness of the neuropil background. Captures out-of-focus background transient patterns.
corr_f_fneu	Trace Stats	<code>corrcoef(F, Fneu)[0, 1]</code>	CRITICAL	Pearson correlation between raw trace and neuropil background. True neurons fire independently from background (low/negative correlation). High correlation (~1.0) indicates background contamination/false detection.
skew_fcorr	Trace Stats	<code>skew(F - 0.7 * Fneu)</code>	Highly Useful	Skewness of neuropil-corrected trace. Extremely clean marker of true, background-corrected calcium transients.

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std_fcorr	Trace Stats	<code>std(F - 0.7 * Fneu)</code>	Highly Useful	Standard deviation of corrected trace. Represents pure calcium transient amplitude.
q10, q25, q50, q75	Percentiles	<code>quantile(Fcorr, p)</code>	Moderate	Lower-to-middle percentiles mapping the baseline quiet state of the corrected trace.
q90, q95, q99	Percentiles	<code>quantile(Fcorr, p)</code>	Highly Useful	Upper percentiles capturing calcium transient firing events. Strong separator of active cells from background noise.
avg_asym	Dynamics	<code>mean(decay / rise)</code>	Highly Useful	Average transient peak asymmetry. Real calcium binding/rise is fast, while decay is slow (asymmetry > 1.5). White noise fluctuations are symmetric (asymmetry ~1.0).
max_asym	Dynamics	<code>max(decay / rise)</code>	Useful	Maximum peak asymmetry. Detects sparsely active cells.
max_width	Dynamics	<code>max(peak_widths)</code>	Useful	Maximum width at half-maximum of detected peaks. Filters out high-frequency noise spikes (width 1-2 frames).
range_fcorr	Amplitudes	<code>max(Fcorr) - min(Fcorr)</code>	Useful	Total peak-to-peak amplitude of background-corrected trace.
range_f	Amplitudes	<code>max(F) - min(F)</code>	Moderate	Total peak-to-peak amplitude of raw trace.
snr	New Biological	<code>std(Fcorr) / std(diff(Fcorr))</code>	Highly Useful	Temporal signal-to-noise ratio. Real neural signals decay slowly and are temporally smooth. Random noise fluctuates rapidly, yielding low SNR.
activity_ratio	New Biological	<code>std(Fcorr > med) / std(Fcorr <= med)</code>	Highly Useful	Ratio of variance above baseline median to variance below. Cleanly captures positive-only calcium events compared to symmetric noise.
peak_density	New Biological	<code>len(peaks_2sig) / len(Fcorr)</code>	Useful	Fires-per-frame density, counting peaks exceeding 2 standard deviations. Helps distinguish active firing somas from inactive/noise ROIs.